PYTHON PROGRAMMING

Case Study Assignment: ATM Transaction Sequence Simulation

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1. Problem Statement

An ATM (Automated Teller Machine) allows users to perform basic banking transactions such as depositing and withdrawing money. This project simulates a sequence of ATM transactions using Python. The program maintains the account balance, accepts multiple transactions, validates transaction amounts, rejects withdrawals when the available balance is insufficient, and records transaction details. Finally, it displays the transaction history and the initial and final account balances.

2. Objectives

The main objectives of the ATM Transaction Sequence Simulation are:

1. To simulate basic ATM transactions such as deposits and withdrawals.
2. To maintain and update the account balance after each successful transaction.
3. To validate transaction amounts and reject invalid amounts.
4. To prevent withdrawals when the available balance is insufficient.
5. To allow users to perform multiple transactions in a single session.
6. To store transaction details, including the transaction type, amount, and status.
7. To display the transaction history, initial balance, and final balance.

3. Assumptions

The following assumptions are considered in this ATM simulation:

1. The initial account balance is ₹ 1000.
2. The user can perform deposits and withdrawals in a single session.
3. The transaction amount must be greater than zero.

4. A withdrawal is successful only when the requested amount does not exceed the available balance.
5. Only the transaction types *deposit* and *withdrawal* are accepted.
6. The program continues accepting transactions until the user enters *no*.
7. All transaction details are stored in a list during the program execution.
8. The program simulates ATM transactions and does not connect to a real bank account.

4. Algorithm

1. Start the program.
2. Initialize the initial balance to ₹ 1000 and set the current balance equal to the initial balance.
3. Create an empty list to store transaction details.
4. Ask the user to enter the transaction type: deposit or withdrawal.
5. Ask the user to enter the transaction amount.
6. Check whether the amount is less than or equal to zero. If it is, reject the transaction as invalid.
7. If the transaction is a withdrawal, check whether the amount exceeds the available balance. If it does, reject the transaction; otherwise, deduct the amount from the balance.
8. If the transaction is a deposit, add the amount to the balance.
9. If the transaction type is invalid, reject the transaction.
10. Store the transaction type, amount, and status in the transaction history.
11. Ask the user whether they want to perform another transaction.
12. Repeat the transaction process if the user enters yes.
13. Display the initial balance, transaction history, and final balance.
14. Stop the program.

5. Pseudocode

```
START
SET initial_balance = 1000
SET balance = initial_balance
CREATE empty list transactions
SET choice = "yes"

WHILE choice = "yes"
    INPUT transaction_type
    INPUT amount

    IF amount <= 0 THEN
        DISPLAY "Invalid amount"
        SET status = "Rejected"
    ELSE IF transaction_type = "withdrawal" THEN
        IF amount > balance THEN
            DISPLAY "Insufficient balance"
```

```
            SET status = "Rejected"
        ELSE
            balance = balance - amount
            DISPLAY "Withdrawal successful"
            SET status = "Successful"
        END IF
    ELSE IF transaction_type = "deposit" THEN
        balance = balance + amount
        DISPLAY "Deposit successful"
        SET status = "Successful"
    ELSE
        DISPLAY "Invalid transaction type"
        SET status = "Rejected"
    END IF

    ADD (transaction_type, amount, status) TO transactions
    INPUT choice
END WHILE

DISPLAY initial_balance
DISPLAY all transactions
DISPLAY balance
STOP
```

6. Python Source Code

ATM-Transaction-Sequence-Simulation.py

```
initial_balance = 1000.00
balance = initial_balance
transactions = []
choice = "yes"

while choice == "yes":
    transaction_type = input("Enter transaction type (withdrawal/deposit): ").lower()
    amount = float(input("Enter amount: "))

    if amount <= 0:
        print("Invalid amount.")
        transactions.append((transaction_type, amount, "Rejected"))
    elif transaction_type == "withdrawal":
        if amount > balance:
            print("Insufficient balance for withdrawal.")
            transactions.append((transaction_type, amount, "Rejected"))
        else:
            balance -= amount
            transactions.append((transaction_type, amount, "Successful"))
            print("Withdrawal successful.")
            print("Remaining balance: ₹", balance)
    elif transaction_type == "deposit":
        balance += amount
        transactions.append((transaction_type, amount, "Successful"))
        print("Deposit successful.")
        print("Updated balance: ₹", balance)
```

```
    else:
        print("Invalid transaction type.")
        transactions.append((transaction_type, amount, "Rejected"))

    choice = input("Do you want to perform another transaction? (yes/no): ").lower()

print("\n--- ATM TRANSACTION SUMMARY ---")
print(f"Initial balance: ₹{initial_balance}")
print("\nTransaction History:")
for transaction in transactions:
    print(transaction)
print("\nFinal balance: ₹", balance)
print("Thank you for using the ATM!")
```

7. Test Cases

S.No	Transaction	Input Amount	Expected Result
1	Withdrawal	₹ 200	Successful; balance becomes ₹ 800
2	Deposit	₹ 500	Successful; balance becomes ₹ 1,300
3	Withdrawal	₹ 5000	Rejected; insufficient balance
4	Deposit	₹ 0	Rejected; invalid amount
5	Invalid type	₹ 100	Rejected; invalid transaction type

8. Output Screenshots

Screenshot 1: Successful Deposit and Withdrawal Transactions

```
PS D:\CODING> C:\Users\ASUS\AppData\Local\Python\bin\python.exe d:/CODING/ATM-
Transaction-Sequence-Simulation.PY
Enter transaction type (withdrawal/deposit): withdrawal
Enter amount: 200
Withdrawal successful.
Remaining balance: ₹ 800.0
Do you want to perform another transaction? (yes/no): yes
Enter transaction type (withdrawal/deposit): deposit
Enter amount: 500
Deposit successful.
Updated balance: 1300.0
Do you want to perform another transaction? (yes/no): no

ATM TRANSACTION SUMMARY
Initial balance: ₹1000.0
Transaction History:
('withdrawal', 200.0, 'Successful')
('deposit', 500.0, 'Successful')
```

```
Final balance: 1300.0
Thank you for using the ATM!
```

Screenshot 2: Rejected Transactions

```
PS D:\CODING> C:\Users\ASUS\AppData\Local\Python\bin\python.exe d:/CODING/ATM-
Transaction-Sequence-Simulation.PY
Enter transaction type (withdrawal/deposit): withdrawal
Enter amount: 5000
Insufficient balance for withdrawal.
Do you want to perform another transaction? (yes/no): no

ATM TRANSACTION SUMMARY
Initial balance: ₹1000.0
Transaction History:
('withdrawal', 5000.0, 'Rejected')
Final balance: ₹1000.0
Thank you for using the ATM!
```

9. Conclusion

The ATM Transaction Sequence Simulation was successfully developed using Python. The program performs deposits and withdrawals, maintains the account balance, validates transaction amounts, and rejects invalid transactions and withdrawals when sufficient funds are not available. It also supports multiple transactions in a single session and stores transaction details in a list. Finally, the program displays the transaction history along with the initial and final balances. This project helped in understanding Python concepts such as variables, conditional statements, loops, lists, tuples, and user input.